Adding and subtracting polynomials

1

a 9 | b 8

2

a 1 term | b 5

3

a Yes, this expression is a monomial.

b The variable is a, the coefficient is -2 and the degree is 3

4

a
- i 2 terms
- ii $-5, -1$

b
- i 3 terms
- ii $1, 7, -1$

5

a Yes | b No | c Yes

6

a False | b True

7

a
- i 7
- ii 2
- iii 2

b
- i 3
- ii $-\frac{7}{10}$
- iii 6

c
- i 7
- ii $\frac{1}{5}$
- iii 5

8

a $2x^2 + 14x - 1$

b $-12x^2 - 8x - 3$

c $-4x^3 - 9x^2 - 17x - 7$

d $\frac{5}{6}x^2 + \frac{3}{20}x + \frac{1}{4}$

e $6.3x^2 - 17.3x + 2.4$

f $-5xy^2 - 3x^2y + 12$

g $-6x^2y^2 = xy^2 - 4y^2$

h $-2x^3 + 6x^2 - 7$

i $-7x^2 - 7x + 26$

j $-9x^3 + 4x^2 + 3x - 1$

k $2x^2 - x - 13$

l $-11x^3 + 9x - 2$

m $12x^2y + xy^2 + 5$

n $-2x^3 - 3x^2 + 9x + 5$

o $\frac{5}{12}x^2 + \frac{7}{6}x - \frac{11}{12}$

p $6.2x^2 + 2.9x + 16.9$

q $3x^3 - 8x - 7$

r $3x^2 + 7x + 1$

9 $12x^3 + 34x^2 - 18x + 6$

10 $2.4q^2 + 59q + 580$

11 $-5.5m^2 + 40.2m + 160$

12 $-x^2 + 3x - 13$

13 $-5x^2 + x - 13$

14 $-7x^2 - 9x + 11$

15 $8x^2 - 10x - 3$

16 $6x^2 + 4x - 1$

17

a equal to 2

b quadratic polynomial

c $7x^2 + 4x - 8 - (5x - 7)$

18

a $2x^3 - 6x^2 - x + 4$

b 3

c 2

d Yes

19

a 6

b -2

20

a 12

b 0

c Yes

21

a $4x^3 - 4x^2 - 10x - 4$

b $-5x^3 - 4x^2 - 10x - 1$

22

a 6

b 6

c 6

23 $b = 12$

24

a $8x^2 + 18x + 14$ units

b $x^2 + 24x$ units

c $4x^3 + 6x^2 + 3x + 8$ units

Additional questions

26

a No

b Yes

c Yes

d No

27

a

i $9{,}5$

ii $4y + 8x + 14$

b

i $6x{,}7x$

ii $13x + 4 - 5x^2$

c

i $4a, -6a$

ii $-2a + 2b - 3ab$

d

i $-4y^3, -7y^3$

ii $4xy - 11y^3 + 9x^3 - 4 + 9x^3y$

28

a No

b No

c Yes

d No

29 Any polynomial with two terms where one term is $2x^4$ and the second term is anything with an exponent less than 4. For example, $2x^4 + 3x$.

30 Any constant term will satisfy the conditions. For example, 5.

31

a Answers may vary.
Any term with a coefficient of 5 and an exponent larger than 3. For example, $5x^4$.

b $-2x^3$

c Answers may vary.
Any term with an exponent of 2 or a constant term. For example, x^2 or 6.

d $-4x^3$

32

a Answers may vary.
The lowest possible degree is 2 if the three exponent blanks are filled with 0, 1, and 2
For example: $3x^2 + 4x^1 - 5x^0 + 6x + 7$ which simplifies to $3x^2 + 10x + 2$.

b Answers may vary.
The highest single digit is 9 so this will be the highest possible leading coefficient.
For example: $9x^5 + 8x^4 - 7x^3 + 6x + 5$.

c Answers may vary.
The minimum number of terms possible is three.
For example: $0x^3 + 4x^2 - 5x^1 + 6x + 7$ which simplifies to $4x^2 + x + 7$.

33

a $(5x + 1) - (2x) = 3x + 1$

b $(5x + 1) + (5x + 1) + (x + 3) + (x + 3) = 12x + 8$

c $(5x^2 + 16x + 3) - (x^2 + 3x) = 4x^2 + 13x + 3$

34 If the three terms in each trinomial have the same exponents, so that there are three pairs of like terms, and one of the pairs of like terms have the same coefficient but with opposite signs, then those two terms will eliminate each other when added.
For example, $x^2 + 2x + 1$ and $x^2 - 2x + 4$ would add together to give $2x^2 + 5$.

35 m

36

a -1 and 1 b 4 and -3 c 4 and 1 d 4, 1 and 1

37 $(8.5 - 2x) + (8.5 - 2x) + (11 - 2x) + (11 - 2x) + x + x + x + x + x + x + x = 39$

38

a Always true. A polynomial is a collection of terms in the form mx^n where m is a real number and n is a nonzero integer. When adding polynomials together, terms are either combined if they have the same value of n or left alone if not. That means all terms in the resulting polynomial will also be a collection of terms in the form mx^n and will also be a polynomial.

b Always true. A linear function can only have a linear term in the form mx and/or a constant term in the form b, both of which are terms in a polynomial.

c Never true. At least one term in the non-polynomial will not combine with any of the terms in the polynomial and such terms will be left in the resulting sum unchanged, which makes the resulting sum a non-polynomial. For example: $(x^2 + \frac{1}{x}) + (3x - 5 = x^2 + 3x - 5 + \frac{1}{x}$

d Sometimes true. If the terms that don't fit the polynomial definition are equal, but with opposite signs, they will eliminate each other and the result will be a polynomial. For example: $(x^2 + \frac{1}{x}) + (3x - \frac{1}{x} = x^2 + 3x$